

Upfitter Relay Box



Principles of Operation

When the ignition is in run or accessory, the upfitter switch assembly receives voltage from the Body Control Module (BCM) . The upfitter switches control the 4 upfitter relays in the upfitter relay box. When the relay is energized, voltage is supplied to the customer installed accessories.

Inspection and Verification

1. Verify the customer concern.
2. Visually inspect for obvious signs of electrical damage.

Visual Inspection Chart

Electrical
• Battery Junction Box (BJB) fuse(s): ▪ 18 (30A) (upfitter relay 1) ▪ 19 (30A) (upfitter relay 2) ▪ 28 (10A) (upfitter relay 4) ▪ 44 (15A) (upfitter relay 3) • Body Control Module (BCM) fuse 41 (7.5A) (upfitter switches) • Wiring, terminals or connectors • Upfitter relay(s) • Upfitter switch

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.
4. If the cause is not visually evident, verify the symptom and GO to [Symptom Chart](#).

Symptom Chart

Symptom Chart		
Condition	Possible Causes	Action
• The upfitter switch controlled component is inoperative	• Fuse • Wiring, terminals or connectors • Customer installed accessory	• GO to Pinpoint Test I.

	- Upfitter relay - Upfitter switch - Body Control Module (BCM)	
An upfitter switch on indicator is inoperative	Upfitter switch	INSTALL a new upfitter switch. REFER to Upfitter Switch in this section. TEST the system for normal operation.

Pinpoint Tests

Pinpoint Test I: The Upfitter Switch Controlled Component is Inoperative

Refer to Wiring Diagrams Cell [140](#) , Customer Access for schematic and connector information.

Normal Operations

With the ignition is in run or accessory, voltage is supplied to the upfitter switches from the Body Control Module (BCM) . When an upfitter switch is activated (moved to the on position), voltage is routed to the coil side of the corresponding upfitter relay.

Each of the upfitter relays (switch side) are supplied voltage at all times from the Battery Junction Box (BJB) . When the upfitter relay is switched activated, the relay energizes and supplies voltage to the customer supplied accessory.

All of the upfitter relays share the same ground circuit.

This pinpoint is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- Customer installed accessory
- Upfitter relay
- Upfitter switch
- BCM

PINPOINT TEST I : THE UPFITTER SWITCH CONTROLLED COMPONENT IS INOPERATIVE

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.	
I1 CHECK THE UPFITTER SYSTEM	
- Ignition ON. - Operate all the upfitter accessories.	
Are all the upfitter accessories inoperative?	
Yes	VERIFY the BCM fuse 41 (7.5A) is OK. If OK, GO to 12 . If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the short circuit.
No	GO to 15 .
I2 CHECK THE UPFITTER SWITCH ON INDICATORS	

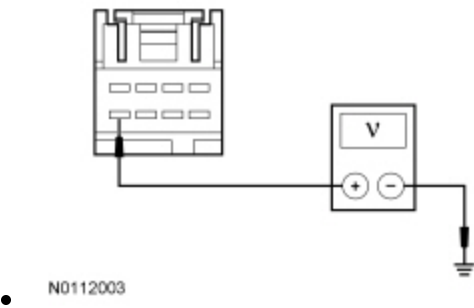
- Place the upfitter switches to the ON position.

Do the on indicators illuminate?

Yes	REPAIR circuit GD138 (BK/WH) between the relays and ground for an open. TEST the system for normal operation.
No	GO to I3 .

I3 CHECK FOR VOLTAGE TO THE UPFITTER SWITCH

- Ignition OFF.
- Disconnect: Upfitter Switch [C3389](#).
- Ignition ON.
- Measure the voltage between the upfitter switch [C3389](#) Pin 8, circuit CBP41 (BU), harness side and ground.

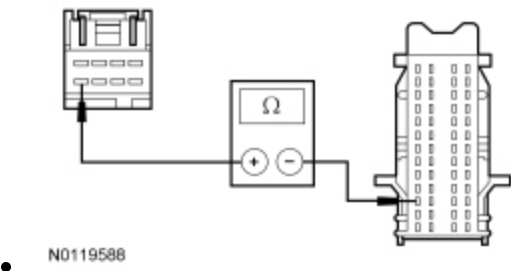


Is the voltage greater than 10 volts?

Yes	INSTALL a new upfitter switch. REFER to Upfitter Switch in this section. TEST the system for normal operation.
No	GO to I4 .

I4 CHECK THE UPFITTER SWITCH VOLTAGE SUPPLY CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: BCM [C2280B](#) .
- Measure the resistance between the upfitter switch [C3389](#) Pin 8, circuit CBP41 (BU), harness side and the BCM [C2280B](#) Pin 3, circuit CBP41 (BU), harness side.



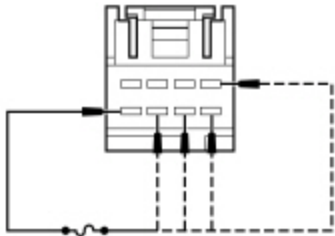
Is the resistance less than 5 ohms?

Yes	GO to I11 .
No	REPAIR circuit CBP41 (BU) for an open. TEST the system for normal operation.

I5 BYPASS THE UPFITTER SWITCH

- Ignition OFF.
- Disconnect: Upfitter Switch [C3389](#).
- Ignition ON.
- Connect a fused jumper wire between the upfitter switch connector, harness side as follows:

Inoperative Switch	Connector-Pin Circuit	Connector-Pin Circuit
Switch 1	C3389 Pin 8 CBP41 (BU)	C3389 Pin 7 CAC01 (GN/BU)
Switch 2	C3389 Pin 8 CBP41 (BU)	C3389 Pin 6 CAC02 (GY/BN)
Switch 3	C3389 Pin 8 CBP41 (BU)	C3389 Pin 5 CAC03 (WH/VT)
Switch 4	C3389 Pin 8 CBP41 (BU)	C3389 Pin 1 CAC04 (WH/BU)



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Does the upfitter accessory operate?

Yes	REMOVE the jumper wire. INSTALL a new upfitter switch. REFER to Upfitter Switch in this section. TEST the system for normal operation.
No	REMOVE the jumper wire. VERIFY the BJB fuse 18 (30A) (relay 1), fuse 19 (30A) (relay 2), fuse 44 (15A) (relay 3) or fuse 28 (10A) (relay 4) is OK. If OK, GO to I6 . If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the short circuit.

I6 CHECK THE UPFITTER RELAY

- Ignition OFF.
- Connect: Upfitter Switch [C3389](#).
- Disconnect: Suspect Upfitter Relay .
- Substitute a known good relay.
- Ignition ON.
- Place the upfitter switch in question to the ON position.

Does the upfitter accessory operate?

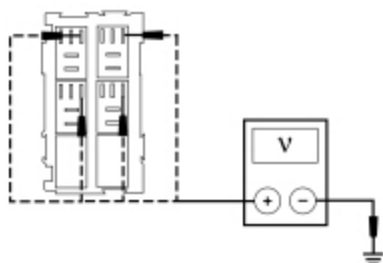
Yes	REMOVE the known good relay. INSTALL a new upfitter relay. TEST the system for normal operation.
No	REMOVE the known good relay. GO to I7 .

I7 CHECK FOR VOLTAGE TO THE UPFITTER RELAY COIL

NOTE: Make sure the upfitter switch and the ignition are on.

- Measure the voltage between the suspect upfitter relay, relay box face side and ground as follows:

Upfitter Relay	Cavity	Circuit
Relay 1	1	CAC01 (GN/BU)
Relay 2	1	CAC02 (GY/BN)
Relay 3	1	CAC03 (WH/VT)
Relay 4	1	CAC04 (WH/BU)



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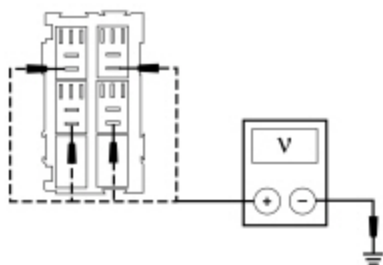
Is the voltage greater than 10 volts?

Yes	GO to I8 .
No	REPAIR the circuit in question for an open. TEST the system for normal operation.

I8 CHECK THE UPFITTER RELAY VOLTAGE SUPPLY CIRCUIT FOR VOLTAGE

- Ignition OFF.
- Measure the voltage between the suspect upfitter relay, relay box face side and ground as follows:

Upfitter Relay	Cavity	Circuit
Relay 1	3	SBB18 (YE/RD)
Relay 2	3	SBB19 (BU/RD)
Relay 3	3	SBB44 (GN/RD)
Relay 4	3	SBB28 (GN/RD)



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Is the voltage greater than 10 volts?

Yes	GO to 19 .
No	REPAIR the voltage supply circuit in question for an open. TEST the system for normal operation.

I9 CHECK THE UPFITTER RELAY GROUND CIRCUIT FOR AN OPEN

- Measure the voltage between the suspect upfitter relay, relay box face side and ground as follows:

Upfitter Relay	Cavity Circuit	Cavity Circuit
Relay 1	3 SBB18 (YE/RD)	2 GD120 (BK/GN)
Relay 2	3 SBB19 (BU/RD)	2 GD120 (BK/GN)
Relay 3	3 SBB44 (GN/RD)	2 GD120 (BK/GN)
Relay 4	3	2

Upfitter Relay	Cavity Circuit	Cavity Circuit
	SBB28 (GN/RD)	GD120 (BK/GN)

Is the voltage greater than 10 volts?

Yes	GO to I10 .
No	REPAIR the upfitter relay ground circuit in question for an open. TEST the system for normal operation.

I10 CHECK THE UPFITTER RELAY OUTPUT CIRCUIT

- Measure the resistance between the upfitter relay in question and the customer access point as follows:

Upfitter Relay	Cavity	Output Circuit
Relay 1	5	CAC05 (YE)
Relay 2	5	CAC06 (GN/BN)
Relay 3	5	CAC07 (VT/GN)
Relay 4	5	CAC08 (BN)

Is the resistance less than 5 ohms?

Yes	The vehicle is operating correctly. REPAIR or INSTALL a new customer installed accessory.
No	REPAIR the upfitter relay output circuit in question for an open. TEST the system for normal operation.

I11 CHECK FOR CORRECT BCM OPERATION

- Disconnect all the BCM connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the BCM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new <u>BCM</u> . REFER to Body Control Module (BCM) in this section. TEST the system for normal operation.
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No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.
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